Membrane Cosmology Foundation Layer: Closed-form Derivation of FIRAS μ-distortion Upper Bound and Universal Density Coupling

— v4.7.8 companion paper (v4.8) —

Shinobu Sakaguchi

Sakaguchi Seimensho (Independent Research), Shisō, Hyogo, Japan

sakaguchi-physics.com

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Abstract

The membrane cosmology v4.7.8 main body (Sakaguchi, 2026a) has established Condition 15 (C15) at A-grade on three independent fronts—SPARC 175 galaxies, dSph 31 galaxies, and HSC-SSP Y3 + GAMA DR4 + KiDS—rejecting MOND at $p = 1.66 \times 10^{-53}$. This companion paper provides the theoretical foundation layer supporting that observational establishment, giving it in closed form and confirming consistency with observations at 2.32% agreement using NGC 3198 as anchor.

There are five core results: (M1) Expanding the FIRAS μ-distortion bound (Fixsen et al., 1996) via the Chluba 2016 W_μ kernel with V_ξ primary, we derive $\alpha_{\rm PT}^{\rm upper} = 2.96 \times 10^{-53}$ (phase-transition coupling upper bound). (M2) Through $\alpha_{\rm PT}^{\rm upper}$ we set a FIRAS-only bound $\tau_m \in [1.9 \times 10^{10},\ 1.4 \times 10^{53}]$ s on the membrane memory time, narrowed to a 5-decade span (38 orders tightened) by the causality constraint from the Local Void ($R = 22$ Mpc). (M3) The SI canonical definition of the mode number $N_{\rm mode}$ (Eq. 2B-II) reproduces the NGC 3198 $k_{\rm mem} \cdot N_{\rm mode}$ product at 2.32% precision (A-grade). (M4) The symbol χ carries two distinct variables χ_E [s^2] and χ_F [m$^{3/2}$/s^2], with no universal conversion between them, stated explicitly. (M5) The Pantheon+ result (Scolnic et al., 2022) $f = 0.379$ does not contribute to the μ kernel (explicit null).

In addition, coupling slopes obtained from SPARC 124 galaxies (after bridge exclusion) and dSph 30 galaxies—$b_\alpha = +0.1084$ vs $+0.1127$—agree to within 0.5% over a 3.92 dex density range, operationally establishing the universal coupling $u_{\rm mem} \cdot \rho_{\rm gal}^{2b_\alpha}$ (C3 internal memo v3, A-grade). This result decides α (linear) versus γ (threshold) model discrimination in favor of α via the Akaike information criterion (ΔAIC $= -2.00$), providing an operational constraint "variation efficiency $\sim$ 11% (amplitude) / $\sim$ 5.5% (power)" that requires no dark-matter particle model. The two values reflect the same $b_\alpha = 0.11$ under two reference-choice conventions (ρ^1 amplitude vs ρ^2 power); the first-principles conversion efficiency is a heuristic pair pending the C3-3 (τ_m estimate) + §7.8 (C3-4 P1–P4 branching ratios) analyses.

Limitations: (i) The SI absolute calibration of the dimensionless $T_m = \sqrt{6}$ cannot be determined internally, leaving a 96-order gap between the σ rest-energy candidate and the membrane coherence candidate. (ii) Determining the absolute value of the membrane memory rate $\Gamma_{\rm actual}$ remains a parallel evaluation across three routes (normal mode / FDT / observational proxy), reducible to an energy-branching problem that requires a choice of $\tau_{\rm relax}$ physics. These limitations are targeted for the next version (21 cm + CMB TT cross-check).

Keywords: membrane cosmology, dark matter alternative, MOND rejection, FIRAS μ-distortion, universal density coupling, C15

1. Introduction

1.1 Observational establishment (v4.7.8 main body)

The v4.7.8 main body (Sakaguchi, 2026a) frames dark-matter phenomena as arising from the folding structure of an elastic membrane rather than from a particle species. Observationally, it established the following: (i) The Condition 15 final form $g_c = 0.584 \cdot \Upsilon_d^{-0.361} \cdot \sqrt{a_0 \cdot v_{\text{flat}}^2/h_R}$ holds on SPARC 175 galaxies with $R^2 = 0.607$, $\Delta\text{AIC} = -14.2$, and LOO-CV stability (Lelli et al., 2016). (ii) The Bernoulli universal relation $G_{\text{Strigari}} = s_0(1 - s_0)\, a_0 = 0.228\, a_0$ is reproduced in two independent samples: dSph 31 galaxies ($0.240\, a_0$, 4% agreement) and SPARC bridge outer 30 points ($0.219\, a_0$, 4% agreement) (McConnachie, 2012; Strigari et al., 2008). (iii) The HSC-SSP Y3 + GAMA DR4 three-field combined analysis (157,338 lenses, 503M pairs) yields $g_c = 2.73 \pm 0.11\, a_0$ independently, with $\Delta\text{AIC}(\text{C15 vs MOND}) = +472$ (22σ). (iv) MOND is rejected at $p = 1.66 \times 10^{-53}$.

1.2 Why a companion paper is needed

The v4.7.8 main body is phenomenologically complete (19 conclusions), but the physical origin of each condition—why g_c takes this functional form, what range τ_m lies in, what α_{PT} means—is only briefly touched upon. This companion paper does not alter the v4.7.8 conclusions; instead, it makes their theoretical foundation explicit as a chain of more than eleven closed-form equations.

The combined material is too large for a single paper (v4.7.8 is already 12 pages). Separate publication is more accessible for both observational and theoretical readerships (separation strategy finalized at Session +6). This companion paper is submitted independently to arXiv as v4.8 and stands non-interferingly with v4.7.8.

1.3 Roadmap

§2 introduces the Q-core 3-judgment set and the Form B potential, taking the Fluctuation-Dissipation relation (Eq. I) for the equilibrium variance of the membrane strain field ϵ as the starting point and chaining four core equations (Eqs. I, II-b, II-d, III, IV; foundation_α1). §3 gives the SI canonical definition of the mode number N_{mode} (Eq. 2B-II) and the symbolic closed form of the UV cutoff Λ_{UV} (Eq. 2C-0; foundation_α2). §4 expands the FIRAS μ-distortion via the Chluba 2016 W_μ kernel and derives $\alpha_{\text{PT}}^{\text{upper}}$ and the τ_m bound. §5 presents the observational cross-checks (NGC 3198 2.32% PASS; universal $b_\alpha = 0.11$ across 3.92 dex). §6 organizes the five main claims M1–M5, and §7 discusses limitations and future work. All numerical anchors and references are collected in the Appendix.

Note to the reader: This paper focuses on tracing the closed-form chain (Eq. I → II-b → II-d → III → IV → 2A-I/II → 2B-I/II → 2C-0 → iv-d-I/II/III/IV → iv-e-I) and entrusts individual derivation details to the accompanying document `foundation_integrated.pdf` (15 pages, with cross-references in Appendix A). The present paper concentrates on making the argument structure explicit; computational reproducibility is ensured by the accompanying scripts (executable via `uv run`, Windows Claude Code environment verified).

1.4 Preview of main claims (M1–M5)

#	Claim	Section	Core number
M1	$\alpha_{\text{PT}}^{\text{upper}} = 2.96 \times 10^{-53}$ (V_ξ primary)	§4	FIRAS $\mu < 9 \times 10^{-5}$
M2	τ_m bound $[1.9 \times 10^{10},\ 1.4 \times 10^{53}]$ s → 5 decades	§4, 5	Local Void $R = 22$ Mpc
M3	Eq. 2B-II SI canonical	§3	NGC 3198 2.32% PASS
M4	χ_E, χ_F independent (no universal conversion)	§3, 7	$[\text{s}^2]$ vs $[\text{m}^{3/2}/\text{s}^2]$
M5	$f = 0.379$ non-contributing to μ kernel (explicit null)	§7	Pantheon+ (Scolnic et al., 2022)

2. Theoretical Framework: Form B + FDT + Picture T

2.1 Q-core 3-judgment set

As the starting point of the entire foundation we adopt three judgments. These are minimal choices required by both internal consistency and observational anchoring; every closed form below presumes all three.

Q-core1 (C-judge): $\chi_E = a_0/k_{\rm mem}$ [s^2].
Adopting the MOND external anchor a_0 avoids circular reference.

Q-core2 (P-T): m_σ^2 (static curvature) and τ_m (dynamic relaxation) combine as an orthogonal, independent product.

Q-core3 (staged adoption): Stage 1 (volume average) → Stage 2 (precision), two stages.

2.2 Form B potential and $T_m = \sqrt{6}$

We adopt the effective potential $U(\epsilon; c) = -\epsilon - \epsilon^2/2 - c\,\ln(1-\epsilon)$ (Form B, established at v4.7.6). Solving $dU/d\epsilon = 0$ gives $\epsilon_0 = \sqrt{1-c}$, and $w(c)^2 = U''(\epsilon_0) = 2\epsilon_0/(1-\epsilon_0)$ guarantees thermodynamic stability over $c \in (0,1)$. At the canonical $c_0 = 0.83$ (SPARC+dSph volume average) we have $\epsilon_0 = 0.412311$ and $w(c)^2 = 1.4032$.

The critical temperature of the Z_2 symmetry-breaking is given dimensionlessly as $T_m = \sqrt{6}$ (Lemma 5; v4.7.6 A-grade (Sakaguchi, 2026c)). This value is a universal constant obtained by combining the Gaussian evaluation of the 4-point correlator with the mean-field critical exponent $z \cdot \nu = 1/2$. Translation to a physical unit (Kelvin equivalent) cannot be decided internally (see §7).

2.3 foundation_α1: the four core equations

From the Langevin dynamics of the membrane strain field, four core equations are obtained in chain (established in Sessions +2 to +6).

$$\begin{aligned}
\text{Eq. I (FDT):}\quad & \langle\delta\epsilon^2\rangle_{\rm eq} = T_m/w(c)^2 = 1.745 \quad (c_0 = 0.83)\\
\text{Eq. II-b:}\quad & m_\sigma^2(c) = 4\epsilon_0 \cdot w(c)^2 \cdot (k_{\rm mem}/a_0)\\
\text{Eq. II-d:}\quad & J_{\rm half}(c) = (8k_{\rm mem}^2 T_m^2 \Gamma/a_0^2) \cdot \epsilon_0(1-\epsilon_0)\\
\text{Eq. III:}\quad & m_\sigma^2 \cdot \langle\delta\epsilon^2\rangle_{\rm eq} = 4\epsilon_0 T_m \cdot (k_{\rm mem}/a_0)\\
\text{Eq. IV:}\quad & \mu = A_{\rm PT} \cdot I_\mu \quad \text{(Stage 1/2 factorization)}
\end{aligned}$$

Eq. III is a structural identity: $w(c)^2$ cancels in the product of Eqs. I and II-b, and the right-hand side reduces to a function of c_0 through ϵ_0 alone. The Jeans volume-halved quantity $J_{\rm half}$ reproduces 30 dSph galaxies in the Phase C2 analysis independently of the Strigari relation ($\pm 5\%$) and functions as an observational anchor of the closed-form system.

Factorization of Eq. IV. The μ-distortion is a direct product of $A_{\rm PT}$ (prefactor) and I_μ (kernel integral):

$$A_{\rm PT} = \frac{4\alpha_{\rm PT} T_m k_{\rm mem} \Gamma N_{\rm mode}}{a_0} \cdot \frac{\rho_{\rm mem,0}^2}{\rho_{\gamma,0}}.$$

I_μ is the integral under the Chluba W_μ kernel, separated into Stage 1 (closed form) and Stage 2 (precision). The ratio S2/S1 = 0.8747 is c_0-independent at machine precision (std = 2×10^{-16}), a strong testament to the correctness of the integrand structure (as shown in §4).

3. foundation_α2: c_{mem}, N_{mode}, and Λ_{UV}

3.1 Step 2-A: membrane sound speed c_{mem} and χ_F

We define a new force-conjugate quantity χ_F that connects the membrane sound speed c_{mem} to χ of Q-core1. The two χ's are easily conflated by symbolic identity but are clearly distinguished by their dimensions (see M4 below).

$$\text{Eq. 2A-I:} \quad k_{\text{mem}}^2 = a_0/c_{\text{mem}}^2$$
$$\text{Eq. 2A-II:} \quad \chi_F = c_{\text{mem}} \cdot \sqrt{a_0} \ \ [\text{m}^{3/2}/\text{s}^2]$$

The numerical evaluation at $c_0 = 0.83$ adopts the $f_{\text{opt}}(c) \cdot v_{\text{flat}}$ mapping of the v3.7 Chap. 18 Table 18-3. From $f_{\text{opt}}(0.83) = 1.9163$ (Sb-type neighborhood extrapolation) and $v_{\text{flat}}^{\text{universal}} = 2.00 \times 10^5$ m/s, we obtain $c_{\text{mem}} = 3.833 \times 10^5$ m/s, $\chi_F = 4.198$ m$^{3/2}$/s^2, and $k_{\text{mem}} = 2.861 \times 10^{-8}$ m^{-1}. Because the extrapolation is from two points, its accuracy is a reserved issue (§7).

3.2 Step 2-B: SI canonical N_{mode} (M3)

The membrane mode density n_{mode} [m^{-3}] and the mode number per unit energy N_{mode} [J^{-1}] are given in terms of Λ_{UV} (UV cutoff) and c_{mem}. The core of M3 is that numerical substitution must always use Eq. 2B-II.

$$\text{Eq. 2B-I:} \quad n_{\text{mode}} = \frac{V \cdot \Lambda_{\text{UV}}^3}{6\pi^2\hbar^3 c_{\text{mem}}^3} \ \ [\text{m}^{-3}]$$
$$\text{Eq. 2B-II:} \quad N_{\text{mode}} = \frac{V \cdot \Lambda_{\text{UV}}^2}{6\pi^2\hbar^3 c_{\text{mem}}^3} \ \ [\text{J}^{-1}] \quad \leftarrow \text{SI canonical}$$

The most stringent cross-check with observation is achieved at NGC 3198. The observed value of the $k_{\text{mem}} \cdot N_{\text{mode}}$ product fixed by C15 is 2.93×10^{38}; the foundation prediction via Eq. 2B-II is 2.998×10^{38}, a relative difference of 2.32% (A-grade, Session +8). This cross-check is the strongest endorsement of the SI canonical nature of Eq. 2B-II and constitutes the empirical content of claim M3.

3.3 Step 2-C: symbolic closed form of Λ_{UV}

For the same N_{mode}, a symbolic closed form Eq. 2C-0 that explicitly includes c_{lt} (speed of light) also exists. It is useful for structural investigations (UV-IR duality, dimensional scans) but fails catastrophically under SI numerical substitution.

$$\text{Eq. 2C-0:} \quad \Lambda_{\text{UV}} = 2\, c_{\text{lt}}^2 \cdot w(c) \cdot \sqrt{\epsilon_0(c)} \cdot c_{\text{mem}}^{-1/2} \cdot a_0^{-1/4}$$

(symbolic; numerical substitution forbidden)

At $c_0 = 0.83$: $\Lambda_{\text{UV}} = 9.549 \times 10^{-49}$ J (v3.7 Table 18-4 linear extrapolation; requires verification due to 2-point extrapolation).

132-order discrepancy. Substituting the SI numerical value at $c_0 = 0.83$ directly into Eq. 2C-II causes c_{lt}^n ($n \sim 20$) to diverge near 10^{170}, producing a discrepancy of 10^{132} against the correctly-signed value via Eq. 2B-II (NGC 3198 consistent value $\sim 10^{38}$). This is a qualitative, not numerical, breakdown, and the role separation between Eq. 2B-II and Eq. 2C-II is a non-retractable rule.

3.4 Stage 2 I_μ integration and the Chluba W_μ kernel

The time integral of the μ-distortion kernel is expanded via the Chluba 2016 $W_\mu(z)$ shape function:

Eq. iv-d-II-a (Chluba W_μ):

$$W_\mu(z) = \left(1 - \exp\left(-(z/Z_\mu)^{1.88}\right)\right) \cdot \exp\left(-(z/Z_{\rm DC})^{2.5}\right)$$
$$Z_\mu = 5 \times 10^4, \quad Z_{\rm DC} = 1.98 \times 10^6$$

Eq. iv-d-I (Stage 1 closed form):

$$I_\mu^{S1}(c_0) = \epsilon_0(c_0) \cdot \frac{\ln((1+Z_{\rm DC})/(1+Z_\mu))}{H_0\sqrt{\Omega_R}}$$

At $c_0 = 0.83$: $I_\mu^{S1} = 7.2275 \times 10^{19}$ s (expected 7.26×10^{19}, 0.45% PASS).
Eq. iv-d-II v2 (Stage 2 integrand):

$$\text{integrand}(z, c_0) = \epsilon_0(c_0) \cdot W_\mu(z) \cdot (1+z)/H(z)$$

That $(1+z)$ appears in the numerator of the Stage 2 integrand is critical. A swap with the denominator produces a systematic error on the order of 30%, whereas the correct placement leaves S2/S1 = 0.8747 independent of c_0 at machine precision (std $= 2 \times 10^{-16}$). This invariance serves as a self-diagnostic of the Stage 2 implementation.

4. Results I: FIRAS μ-distortion and $\alpha_{\rm PT}^{\rm upper}$ (M1, M2)

4.1 Closed form of $B_{\rm FIRAS,combo}$

Connecting the FIRAS μ upper bound (Fixsen et al., 1996) $\mu_{\rm FIRAS} < 9 \times 10^{-5}$ to the $A_{\rm PT}$ prefactor of Eq. IV yields the following closed form:

Eq. iv-d-III:

$$B_{\rm FIRAS,combo}(c_0) = \frac{\mu_{\rm FIRAS} \cdot A_0 \cdot \rho_{\gamma,0}}{4\, T_m\, \Gamma_{\rm ref}\, \rho_{\rm mem,0}^2\, I_\mu(c_0)}$$

At $c_0 = 0.83$: $B_{\rm FIRAS,combo} = 1.516 \times 10^{-12}$ [J^{-1} m$^{-1/2}$].

A1 operational working hypothesis (inherited #1–17). The numerical anchors of this section (in particular $\rho_{\rm mem,0} = 2.3 \times 10^{-27}$ kg/m^3) presuppose the v4.7.8 main-body assumption $\rho_{\rm mem,0} = \rho_{\rm DM,0}$ (operational). A1 is a working hypothesis at the observational-establishment stage; the present foundation guarantees the consistency of closed forms under A1.

We unify SI units with $\rho_{\rm mem,0} = 2.3 \times 10^{-27}$ kg/m^3 (A1) and $\rho_{\gamma,0} = 4.6 \times 10^{-31}$ kg/m^3. $\Gamma_{\rm ref} = 1$ is a placeholder; the linear scaling rule (see §6) guarantees that the actual $\Gamma_{\rm actual}$ can be applied a posteriori.

4.2 Derivation of $\alpha_{\rm PT}^{\rm upper}$ (M1)

The closed form Eq. iv-d-IV that provides an upper bound on $\alpha_{\rm PT}$ from $B_{\rm FIRAS,combo}$ becomes a $c_{\rm mem}$-linear contracted form (after V_ξ adoption). The sensitivity to the mode-volume choice V is shown below:

V choice	V [m^3]	$\alpha_{\rm PT}^{\rm upper}$	Interpretation
V_ξ (membrane coherence)	7.683×10^{63}	2.96×10^{-53}	primary (consistent with a_0 independence)
$V_{\rm cosmo}$ (geometric)	1.060×10^{79}	2.15×10^{-68}	conservative upper
Ratio $V_{\rm cosmo}/V_\xi$	$\sim 1.38 \times 10^{15}$	$\sim 1.38 \times 10^{15}$	linear scaling in V

Adopting V_ξ as primary is the only choice consistent with the "a_0 independence" structural finding of foundation_α2. $V_{\rm cosmo}$ includes geometric volumes above galaxy-cluster scales and thus overestimates physical coherence.

M1 claim: $\alpha_{\rm PT}^{\rm upper} = 2.96 \times 10^{-53}$ (V_ξ primary, $c_0 = 0.83$).

4.3 τ_m bound and Local Void narrowing (M2)

Inverting through $\alpha_{\rm PT}^{\rm upper}$ onto the membrane memory time τ_m, we obtain two ends —a FIRAS-only upper bound and a causality lower bound:

$$\begin{aligned} \tau_m^{\rm upper}\ (V_\xi, \text{FIRAS-only}) &= 1.418 \times 10^{53}\ \text{s} \\ \tau_m^{\rm lower}\ (\text{causal},\ z = 5 \times 10^4\ \text{Hubble}) &= 1.906 \times 10^{10}\ \text{s}\ \ (\sim 605\ \text{yr}) \\ &\Rightarrow\ \text{FIRAS-only span} = 40\text{–}58\ \text{decades} \end{aligned}$$

This span can be narrowed by the causality constraint of the P2 void. We define the void-interior τ_m upper bound as the minimum of two routes via Eq. iv-e-I:

Eq. iv-e-I: $B_{\tau_m,\rm void} = R_{\rm void}/v_{\rm char}$ (min of 2 routes).
Route 1 (causality): $B_{\rm causal} = R_{\rm void}/c_{\rm lt}$
Route 2 (dynamical): $B_{\rm dyn} = R_{\rm void}/v_{\rm flat,void,median}$
$\tau_m^{\rm upper,final} = \min(B_{\tau_m,\rm void}, \tau_{\rm FIRAS})$

$V_{\xi,\rm void}$ **primary selection (#27).** Within a void, the mode volume has two candidates: $V_{\xi,\rm void}$ (membrane coherence volume, primary) and $V_{\rm void}$ (geometric volume, conservative upper). This paper adopts $V_{\xi,\rm void}$ as primary to maintain consistency with the V_ξ-primary policy of §4.2. This selection preserves consistency with both the a_0-independence structural finding of foundation_α2 and the void narrowing.

At the Local Void ($R = 22$ Mpc) (Kreckel et al., 2011) we obtain $B_{\rm causal} \sim 2.26 \times 10^{15}$ s; using this as an upper end, $\tau_m \in [1.9 \times 10^{10},\ \sim 2 \times 10^{15}]$ s, compressed to 5 decades (38 orders tighter). Bootes void ($R = 55$) and Corona Borealis void ($R = 60$) give looser bounds than the Local Void.

The $R_{\rm void}$ values of the five-void sample (Local / Bootes / Cetus / Sculptor / Corona Borealis) in this section include design estimates as of Session +15; they will be finalized after literature verification on the Windows side (Kreckel et al., 2011; Pan et al., 2012; Ricciardelli et al., 2014). $\rho_{\rm void}/\bar{\rho}$ and $c_{\rm void,median}$ are assumed to have 30–50% uncertainty.

4.3.1 Tentative numerical values for the 5-void sample

void	R [Mpc]	$B_{\rm causal}$ [s]	source (tentative)	note
Local Void	22	2.26×10^{15}	Kreckel et al. (2011)	tightest
Sculptor Void	25	2.57×10^{15}	Pan et al. (2012)	SDSS DR7 catalog
Cetus Void	30	3.09×10^{15}	Pan et al. (2012)	SDSS DR7 catalog
Bootes Void	55	5.66×10^{15}	Kirshner+1981	classical void
Corona Borealis	60	6.17×10^{15}	Ricciardelli et al. (2014)	largest

The Local Void gives the tightest upper bound; at $R = 22$ Mpc it lies closest to our Galaxy, so the causality constraint is most effective. The more distant voids have longer propagation times $B_{\rm causal} = R/c_{\rm lt}$ and hence provide only looser bounds. The M2 narrowing claim relies on the Local Void; the measurement precision of its R (Kreckel et al., 2011) is the dominant systematic source of the final value.

M2 claim. The bound $\tau_m \in [1.9 \times 10^{10},\ 1.4 \times 10^{53}]$ s (FIRAS-only) narrows to the 5-decade span $[1.9 \times 10^{10},\ 2 \times 10^{15}]$ s under the Local Void causality constraint. A 5-decade span is observationally meaningful and is cross-checkable against future probes (21 cm, CMB TT).

5. Results II: universal density coupling b_α and Γ_{actual}

5.1 C3 foundation: α vs γ model comparison

In C3-2 we took as candidates for the ρ_{gal} dependence of u_{mem}: (a) linear coupling $F = \lambda\,\rho_{\text{gal}}$ (α model), and (b) threshold coupling $F = \lambda\,\rho_{\text{gal}} - 2\,\epsilon_\star$ (γ model). We compare the two directly on SPARC and dSph.

5.2 SPARC-only analysis ($N = 124$)

From SPARC 175 galaxies, with a quality flag $Q < 3$, $v_{\text{flat}} > 0$, and finite g_c^{C15}, and excluding the four bridge galaxies by the Phase C2 protocol (ESO444-G084, NGC2915, NGC1705, NGC3741), we obtain a primary sample of $N = 124$. The bridge exclusion is justified by Lesson 91 (bridge / extreme-regime pre-cut protocol): it averts a spurious $\rho = +0.85$ driven by NGC3741 alone. After exclusion the true signal is partial $\rho = +0.522$ $[+0.369, +0.645]$, $\sigma = 6.37$, ΔAIC vs null $= +46.8$ (A-grade).

In a direct α vs γ comparison, the partial NLL difference is 0.001 (machine precision), and the partial Spearman ρ's are exactly equal ($0.5496 = 0.5496$). The optimal λ_p for γ saturates at the grid upper end (10,000), placing all SPARC 124 galaxies above threshold and mathematically converging $\gamma \to \alpha$. With the +1 parameter AIC penalty this yields $\Delta\text{AIC}(\alpha - \gamma) = -2.00$, favoring α by parsimony (Lesson 92: *parsimony first*).

5.3 dSph extension and 3.92 dex universal coupling

Within the SPARC density range of 2 dex, α vs γ is indiscriminable; dSph is 3–4 dex lower in ρ_{gal} than SPARC and differentially tests the existence of a threshold structure. For the dSph 30 galaxies based on McConnachie (2012) (Sgr excluded, $\Upsilon_V = 1$):

$$\begin{aligned} \text{partial Spearman } \rho &= +0.557, \quad \sigma = 3.27 \\ b_\alpha \text{ (dSph)} &= +0.1127 \\ \Delta\text{AIC}(\alpha - \gamma) &= -2.00 \;\; (\alpha \text{ favored, SCE delta sensitivity robust}) \end{aligned}$$

The SPARC+dSph combined $N = 154$ at 3.92 dex density range yields $\Delta\text{AIC}(\alpha - \gamma) = -1.76$ (LRT $p = 0.31$); no γ threshold is detected. The most important finding is the numerical agreement of slopes between the two independent samples:

sample	N	b_α (partial)	note
SPARC (bridge excluded)	124	+0.1084	g_c^{C15} anchor
dSph (Sgr excluded)	30	+0.1127	G_{Strigari} anchor
|diff|	–	0.0042 (within 0.5%)	across 3.92 dex

Lesson 93 (universal coupling = slope agreement). The fact that slopes independently fit under different physical regimes (SPARC g_c^{C15} vs dSph G_{Strigari}) agree within 0.5% over 3.92 dex meets the operational definition of the universal coupling $u_{\text{mem}} \propto \rho_{\text{gal}}^2$. The key point of this section is that the evidence is the *numerical agreement of slopes*, not the statistical significance of a correlation.

5.3.1 Phase C3 methodology triptych (Lessons 91–93)

The three lessons newly established in Phase C3 Step 2-3 are methodological principles applicable to general observational-data analyses and will be retained in future extensions (UFD extension, void galaxy morphology samples, etc.):

#	Proposition	Operational content
91	bridge / extreme-regime pre-cut protocol	Prior exclusion on the grounds of physical-regime uniformity. SPARC 4 bridge galaxies; dSph Sgr.
92	parsimony first	Information criteria (AIC/BIC) take priority over Spearman strength; model-selection discipline.
93	universal coupling = slope agreement	Numerical slope agreement (within 1%) between two independent samples is the operational definition of universality.

These three lessons provide the evidentiary standard for connecting the closed-form system to observations within the foundation. A significant correlation alone (e.g. $\rho > 0$) is a necessary but not sufficient condition for universality; the stronger requirement of numerical slope agreement gives the first operational empirical warrant for a foundation hypothesis.

5.4 Parallel evaluation of Γ_{actual} across three routes

We attempt to estimate the absolute value of Γ_{actual} from the universal $b_\alpha = 0.11 \pm 0.005$. We evaluate three routes in parallel and ensure internal consistency (absolute-value claims by a single route are withheld):

route	expression	result	units	assessment
A: normal mode	$(1 - b_\alpha)/b_\alpha \cdot \text{mode_factor}$	8.091	dim.less	strong candidate
B: FDT relax.	$\chi_F/(V''\tau_{\text{relax}})$	$4\times10^{-18} \sim 10^{-10}$	$\text{m}^{3/2}/\text{s}$	4 τ_{relax} candidates
C: obs. proxy	SPARC memory > 10 Gyr	$< 5.75\times10^{-18}$	$\text{m}^{3/2}/\text{s}$	upper only

Route A interprets $b_\alpha = 0.11$ as the $u_{\text{mem}} \to$ stress conversion efficiency (an operational expression of the C3-4 energy branching). $\Gamma_A = 8.091$ is a naive representation of this branching (11% observable : 89% dissipation / gravitational wave / internal conversion). Route B shows a 7-decade spread over four τ_{relax} candidates (Hubble / galaxy dynamical / kpc crossing / BE thermalization), leaving the τ_{relax} choice unresolved (§7). Route C provides only a conservative upper bound from SPARC memory > 10 Gyr.

5.4.1 Three-route integrated assessment and integrity

The parallel evaluation across three routes is the core of non-retractable item #30 and reflects the foundation's inherent skepticism toward absolute-value claims from a single route. Route A is derived from the observational grounding of the universal $b_\alpha = 0.11$ and therefore carries more direct evidence than the other two. The 7-decade spread of Route B reflects the ambiguity of the τ_{relax} physical choice, and Route C provides only a conservative upper bound.

As an internal consistency check, normalizing Route A (dimensionless 8.091) and Route B (Hubble-time adoption 4.12×10^{-18} $\text{m}^{3/2}/\text{s}$) by the foundation scale ($\chi_F \cdot V'' \cdot T_m/\hbar$, non-retractable #29) shows a hint of order-of-magnitude agreement (a quantitative determination awaits the C3-4 energy-branching analysis). This internal consistency sets a future-work direction for determining the absolute value of Γ_{actual} (§7.2).

5.4.2 Two interpretations of $b_\alpha = 0.11$ (11% vs 5.5% unification, #17 treatment)

The v4.7.8 non-retractable #17 is "11% / 5.5% treatment", giving two operational interpretations for the same $b_\alpha = 0.11$:

interpretation	basis	efficiency formula	value
(i) memo v3 p.6	ρ^1 (amplitude)	$b_\alpha/1$	11%
(ii) §7.6 p.3	ρ^2 (power, u_{mem} source)	$b_\alpha/2$	5.5%
essence of difference	amplitude vs power	one sqrt step	factor of 2

Both interpretations reference the same $b_\alpha = 0.11$ under different convention choices and are not numerically inconsistent. Against the path-A prescription $u_{\rm mem} \propto F^2 \propto \rho_{\rm gal}^2$, $b_\alpha/2 = 5.5\%$ represents the source power fraction while $b_\alpha/1 = 11\%$ represents the residual-g amplitude fraction. Both are heuristic; the first-principles $u_{\rm mem} \to$ stress $\to g_{\rm obs}$ conversion efficiency will be fixed in the v4.9 phase through C3-3 (τ_m physical estimate) and §7.8 (C3-4 P1–P4 branching ratios, the energy-branching-ratio section).

Accordingly, Route A keeps two versions of its expression:

- $\Gamma_A = 8.091$ (amplitude basis, $(1 - b_\alpha)/b_\alpha = 0.89/0.11$)
- $\Gamma'_A = 17.18$ (power basis, $(1 - b_\alpha/2)/(b_\alpha/2) = 0.945/0.055$)

These are different representations of the same phenomenon and are not numerically inconsistent.

6. Discussion: M4, M5, and overall consistency

6.1 Independence of the χ dual (M4)

The symbol χ refers to two independent variables in the present foundation. The χ_E of Q-core1 (originating in the v4.2 supplementary section A; dimension [s^2]) derives from the MOND anchor $a_0/k_{\rm mem}$, while the χ_F of Eq. 2A-II (dimension [m$^{3/2}$/s^2]) derives from the membrane sound speed $c_{\rm mem}\sqrt{a_0}$. Their dimensional ratio [m$^{3/2}$/s^2]/[s^2] = [m$^{3/2}$/s^4] has no universal physical meaning, and the two χ's must be carefully distinguished throughout the paper. This companion paper acknowledges the absence of any universal conversion positively under the banner M4.

6.2 $f = 0.379$ explicit null (M5)

The dark-component fraction $f = 0.379 \pm 0.029$ as independently measured by Pantheon+ (Scolnic et al., 2022) is a cosmological quantity independent of $c_0 = 0.83$ (SPARC+dSph volume average). In particular it does not contribute to the μ-distortion kernel of this foundation (no f appears in the integrand). By stating this separation explicitly as an M5 null, we prevent reader confusion (a spurious relation $f \sim 1 - c_0$, or an assumed f dependence of the μ kernel).

6.3 Linear scaling rule and modularity

The linear scaling rule against the placeholder $\Gamma_{\rm ref} = 1$ (established in Session +10, non-retractable) guarantees modularity of the entire foundation:

quantity	$\Gamma_{\rm actual}$ dependence	note
$\tau_{\rm from_firas}$	proportional to $\Gamma_{\rm actual}$	linear
$\alpha_{\rm PT}^{\rm upper}$	invariant	cancelled through $B_{\rm FIRAS,combo}$
$B_{\rm FIRAS,combo}$	proportional to $1/\Gamma_{\rm actual}$	inverse
$\tau_m^{\rm upper}$ (FIRAS-only)	proportional to $\Gamma_{\rm actual}$	via FIRAS
$\tau_m^{\rm upper}$ (void-bound)	invariant	purely geometric

This table gives an explicit recipe for how the numerical results of the present paper will update once the absolute value of $\Gamma_{\rm actual}$ (discussed in §7) is established. M1 ($\alpha_{\rm PT}^{\rm upper}$) is invariant, while M2 (the FIRAS-only upper bound on τ_m) is subject to linear scaling.

6.4 Stage 2 self-diagnostic and S2/S1 = 0.8747

The Stage 1 / Stage 2 ratio S2/S1 = 0.8747 being independent of c_0 (std = 2×10^{-16}, machine precision) is strong evidence that the $(1 + z)$ numerator placement in the Eq. iv-d-II v2 integrand is correct. Interchanging the numerator and denominator would produce a c_0-dependence at std $\sim$ 30%. This invariance serves as a self-diagnostic for the Stage 2 implementation and remains a regression test for future numerical re-evaluations.

6.5 Non-interference with the v4.7.8 main body

aspect	v4.7.8 main body	v4.8 foundation
layer	observational establishment	theoretical foundation
main conclusion	C15, MOND rejection $p = 1.66 \times 10^{-53}$	$\alpha_{\mathrm{PT}}^{\mathrm{upper}}$, τ_m bound
statistical power	SPARC 175 + dSph 31 + HSC 503M	NGC 3198 2.32% + Lessons 91–93
arXiv category	astro-ph.CO	astro-ph.CO + hep-th cross
mutual dependence	–	presupposes v4.7.8 results (unchanged)

This companion does not alter the conclusions of v4.7.8. As a purely additive layer, it provides theoretical foundation to the observational establishment of v4.7.8.

7. Limitations and Future Work

7.1 The 96-order gap in SI absolute calibration of T_m

In translating $T_m = \sqrt{6}$ to a physical unit (Kelvin equivalent), a 96-order gap persists between the σ rest-energy route (a) and the membrane coherence route (b). Route (a) gives $\hbar \cdot \Omega_\sigma \sim 10^{-45}$ J (micro scale), while route (b) gives $\rho_{\mathrm{mem},0} \cdot V_\xi \cdot c_{\mathrm{lt}}^2 \sim 10^{47}$ J (macro scale). These are hierarchically distinct physical quantities—individual quantum excitation versus collective mode excitation—and cannot be decided internally. Simultaneous use of external anchors (FIRAS + CMB TT + 21 cm) is the top priority for the next version.

7.2 Determining the absolute value of Γ_{actual} + unifying 11%/5.5% first-principles efficiency

The dimensional consistency between Route A (dimensionless 8.091 / 17.18) and Route B (SI $4 \times 10^{-18} \sim 10^{-10}$ $\mathrm{m}^{3/2}/\mathrm{s}$ with four candidates) is unresolved. We adopt normalization by the foundation scale ($\chi_F V'' T_m/\hbar$) as primary (non-retractable #29), but resolution of the τ_{relax} choice in Route B requires addressing the C3-4 energy-branching problem. Route A's neighborhood is judged a strong candidate, but an absolute-value claim is withheld.

In addition, the two heuristic interpretations written together in §5.4.2—11% (amplitude basis) and 5.5% (power basis)—should be consolidated into a single first-principles $u_{\mathrm{mem}} \to$ stress $\to g_{\mathrm{obs}}$ conversion efficiency. This will be fixed jointly in the v4.9 phase through completion of C3-3 (τ_m physical estimate) and §7.8 (C3-4 P1–P4 branching ratios). These two tasks (the absolute value of Γ_{actual} and the 11%/5.5% unification) are treated as the same pending item.

7.3 Extrapolation accuracy of Λ_{UV} and c_{mem}

$c_{\mathrm{mem}}(0.83) = 3.833 \times 10^5$ m/s is an Sb-type neighborhood extrapolation from two points of the v3.7 Chap. 18 Table 18-3 (range $c \in [0.3, 0.8]$); physical verification outside the range (at 0.83) is incomplete. Similarly, $\Lambda_{\mathrm{UV}}(0.83) = 9.549 \times 10^{-49}$ J is a linear extrapolation from Table 18-4. Improving both extrapolations requires additional samples of $c_{\mathrm{mem}}(c)$ and independent measurements of $\Lambda_{\mathrm{UV}}(c)$ (precision CMB spectral distortion measurements, PIXIE (Kogut et al., 2011)).

7.4 Void sample literature consistency

The R_{void} values for the five voids used in the M2 narrowing (Local / Bootes / Cetus / Sculptor / Corona Borealis) include tentative figures and will be fixed in a formal version via direct input from Kreckel et al. (2011); Pan et al. (2012); Ricciardelli et al. (2014) through Windows implementation task (a) (see Appendix A). $\rho_{\mathrm{void}}/\bar{\rho}$ and $c_{\mathrm{void,median}}$ are assumed to have 30–50% uncertainty; the final version will add void-by-void uncertainty propagation.

7.5 UFD extension (C3 next version)

In the Phase C2 strict 15 subset, 7/15 are below threshold and a marginal γ hint remains with $\Delta\mathrm{AIC} = +0.17$ (a boundary case of Lesson 93). With $N = 15$ the matter cannot be decided, and

securing statistical power through an ultra-faint dwarf (UFD) extension (Segue I/II, Reticulum II, Boötes II) is the observational priority of the next version. UFDs may allow judging the existence of a threshold at density regions 1–2 dex below dSph.

7.6 v4.9 next-version roadmap

priority	issue	external input	expectation
1	T_m SI absolute calibration (resolving the 96-order gap)	FIRAS + CMB TT + 21 cm	decide route (a)/(b)
2	Γ_{actual} absolute value + 11%/5.5% unification	C3-4 energy-branching analysis	Route A/B consistency
3	void literature consistency (finalizing M2 narrow)	Kreckel et al. (2011); Pan et al. (2012)	formal 5-decade span
4	$c_{\text{mem}}, \Lambda_{\text{UV}}$ extrapolation accuracy	precision Chap. 18 Table for $c \in (0.8, 1)$	physical verification
5	UFD extension (C3-2 discrimination)	Segue I/II, Ret II, Boö II	threshold reality

8. Conclusions

This companion paper (v4.8) makes explicit the theoretical foundation layer of the membrane cosmology v4.7.8 main body (Sakaguchi, 2026a) as a chain of more than eleven closed-form equations. The results are summarized as five main claims plus the universal density coupling:

- **(M1)** FIRAS μ-distortion (Fixsen et al., 1996) + Chluba 2016 W_μ kernel + V_ξ primary gives $\alpha_{\text{PT}}^{\text{upper}} = 2.96 \times 10^{-53}$ ($c_0 = 0.83$).
- **(M2)** The τ_m bound $[1.9 \times 10^{10},\ 1.4 \times 10^{53}]$ s (FIRAS-only, 40–58 decades) narrows to 5 decades (38 orders tighter) under the Local Void ($R = 22$ Mpc) causality constraint of Eq. iv-e-I.
- **(M3)** The SI canonical definition of N_{mode} (Eq. 2B-II) reproduces the $k_{\text{mem}} \cdot N_{\text{mode}}$ product at NGC 3198 at 2.32% precision (A-grade).
- **(M4)** The two variables carried by the symbol χ, χ_E [s^2] vs χ_F [m$^{3/2}$/s^2], are independent, and no universal conversion exists.
- **(M5)** $f = 0.379$ from Pantheon+ (Scolnic et al., 2022) does not contribute to the μ kernel (explicit null).

As an independent finding, the coupling slopes from SPARC 124 galaxies (bridge excluded) and dSph 30 galaxies, $b_\alpha = +0.1084$ vs $+0.1127$, agree within 0.5% over the 3.92 dex density range, operationally establishing the universal coupling (C3 memo v3, A-grade; Lessons 91–93). This rejects the γ (threshold) model at $\Delta\text{AIC} = -2.00$ and provides the operational constraint "$u_{\text{mem}} \to g_{\text{obs}}$ variation efficiency $\sim$ 11% (amplitude) / $\sim$ 5.5% (power)" that requires no dark-matter particle model. The two values coexist as different convention bases for the same $b_\alpha = 0.11$ (ρ^1 vs ρ^2); the first-principles conversion efficiency will be fixed in the v4.9 phase (C3-3 + §7.8).

The verification path of this foundation is already supported by two independent tests: the NGC 3198 2.32% PASS (the strongest anchor of the N_{mode} closed form) and the c_0 independence of S2/S1 = 0.8747 (a self-diagnostic for the Stage 2 implementation). Three limitations remain (§7): the 96-order gap in T_m SI calibration, the absolute value of Γ_{actual}, and the first-principles unification of 11%/5.5%. These will be addressed jointly in the v4.9 next version using external anchors (FIRAS + CMB TT + 21 cm).

The accompanying materials of this paper (`foundation_integrated.pdf`, 15 pages, 13 build scripts, `c3_3c_p2_void_template.csv`, `foundation_gamma_actual.py`) are reproducibly executable via `uv run` in the Windows Claude Code environment and will be released on the GitHub Public repo (MIT License) (Sakaguchi, 2026b). Cross-references for all non-retractable items #1–31 are collected in Appendix A of `foundation_integrated.pdf`.

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A. Numerical anchor quick-reference ($c_0 = 0.83$ canonical)

Quantity	Value	Source
c_0 (SPARC+dSph vol avg)	0.83	v4.7.8 §6
$\epsilon_0(0.83) = \sqrt{1-c}$	0.412311	Form B
$w(0.83)^2$	1.4032	Form B
T_m (dimensionless)	$\sqrt{6} = 2.449$	Lemma 5 (Sakaguchi, 2026c)
$c_{\mathrm{mem}}(0.83)$	3.833×10^5 m/s	v3.7 Table 18-3 extrapolation
$\chi_F(0.83)$	$4.198\ \mathrm{m}^{3/2}/\mathrm{s}^2$	Eq. 2A-II
$V''(\phi_0, x = 0)$	2.31	Chap. 18 Q1-α
$\Lambda_{\mathrm{UV}}(0.83)$	9.549×10^{-49} J	Table 18-4 extrapolation
$V_\xi(0.83)$	$7.683 \times 10^{63}\ \mathrm{m}^3$	coherence volume
$B_{\mathrm{FIRAS,combo}}$	$1.516 \times 10^{-12}\ [\mathrm{J}^{-1}\mathrm{m}^{-1/2}]$	Eq. iv-d-III
$I_\mu^{S1}(0.83)$	7.2275×10^{19} s (0.45% PASS)	Eq. iv-d-I
S2/S1 (c_0 independent)	0.8747 (std $= 2 \times 10^{-16}$)	Eq. iv-d-II v2
$\alpha_{\mathrm{PT}}^{\mathrm{upper}}$ (V_ξ)	2.96×10^{-53}	M1
τ_m^{upper} (FIRAS)	1.418×10^{53} s	V_ξ primary
τ_m^{upper} (Local Void)	$\sim 2.26 \times 10^{15}$ s	M2 narrow, 5 decades
τ_m^{lower} (causal $z = 5 \times 10^4$)	1.906×10^{10} s (605 yr)	causality
NGC 3198 $k_{\mathrm{mem}} \cdot N_{\mathrm{mode}}$	2.32% PASS	M3, Eq. 2B-II verification
b_α (universal)	$+0.11 \pm 0.005$	across 3.92 dex
Γ_A (amplitude, dimensionless)	8.091	Route A (11% version)
Γ'_A (power, dimensionless)	17.18	Route A' (5.5% version)
Γ_B (FDT, 4 candidates)	$4.12 \times 10^{-18} \sim 9.58 \times 10^{-11}\ \mathrm{m}^{3/2}/\mathrm{s}$	Route B
Γ_C upper	$< 5.75 \times 10^{-18}\ \mathrm{m}^{3/2}/\mathrm{s}$	Route C (SPARC memory)
f (Pantheon+)	0.379 ± 0.029	M5, μ-kernel non-contributing

B. Cross-reference of non-retractable items #1–31

Range	Origin session	Main content	Reference in this paper
#1–17	inherited from v4.7.8	$\kappa = 0$, $T_m = \sqrt{6}$, Form B, Q-core 3, Eqs. I–IV, 11%/5.5% treatment	§2, §5.4.2
#18–21	C3 memo v3	universal $b_\alpha = 0.11$, Lessons 91–93	§5.2–5.3.1, Abstract
#22–25	Session +11	χ dual, 2B-II/2C-II, dimensionless T_m, eps_scale (a)(b)	§6.1, §3.2–3.3, §7.1
#26–28	Session +13	Eq. iv-e-I 2 routes, $V_{\xi,\mathrm{void}}$, min structure	§4.3
#29–31	Session +14	dimensionless foundation scale, 3 routes, b_α direct	§5.4